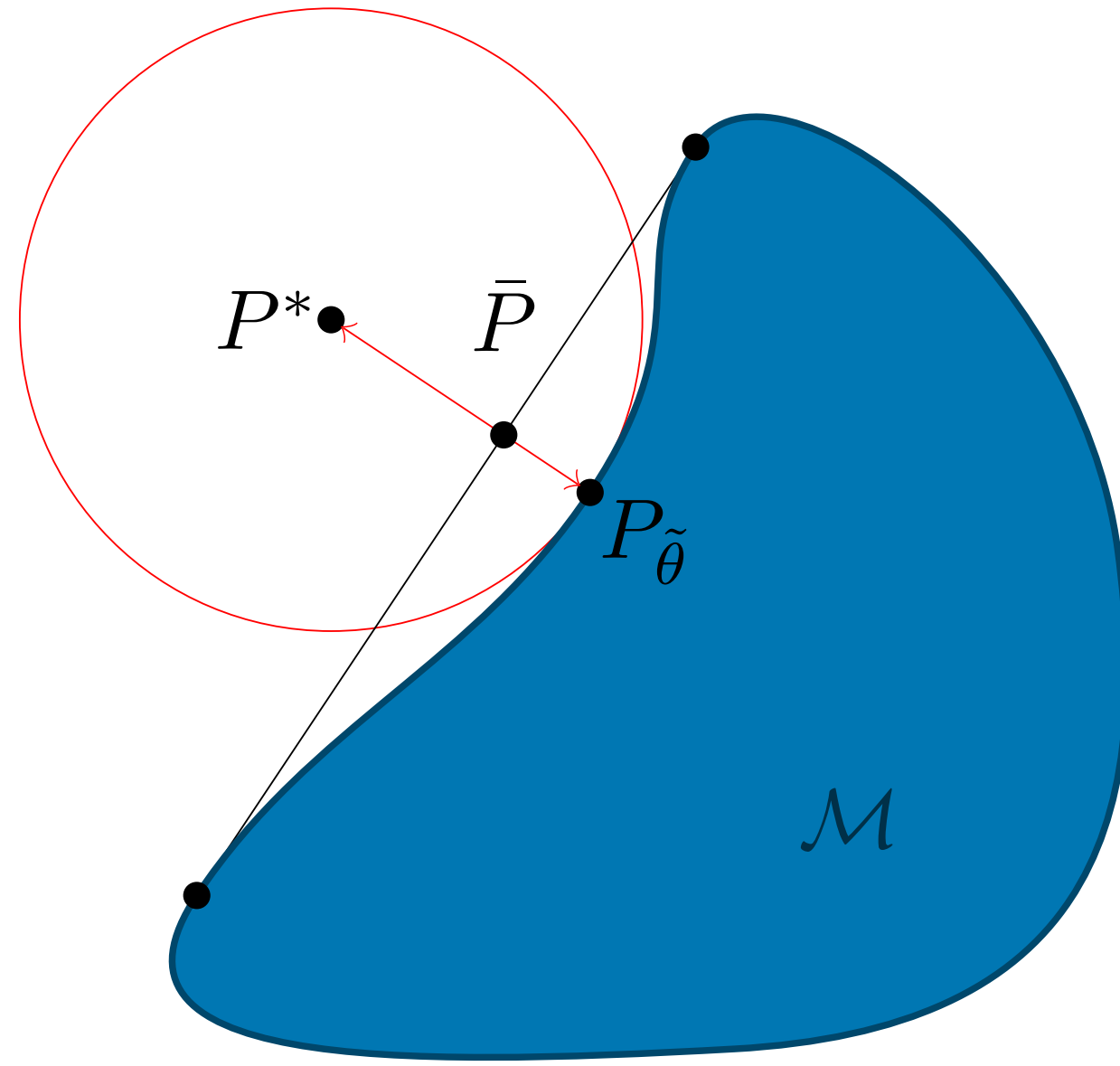


Generalized Bayes I – Learning Rate

Problem: *Bad* misspecification

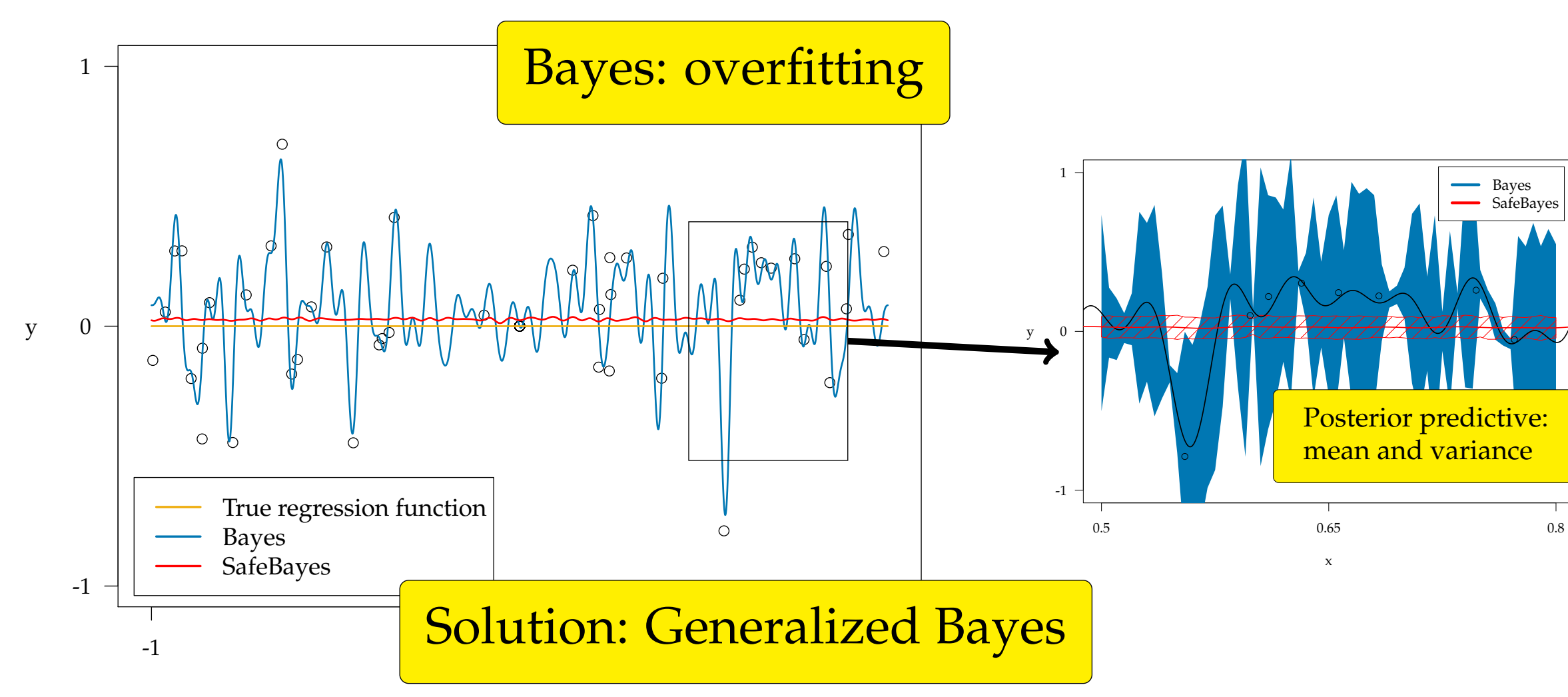
$P_{\hat{\theta}}$ is the closest distribution in the model $\mathcal{M}$ to the true P^* in KL-divergence. When the model is not convex, the posterior predictive distribution $\bar{P}$ might be a mixture of *bad* distributions in the model that ends up outside $\mathcal{M}$. We get:



- Bad square-risk behaviour
- Good log-risk behaviour

This discrepancy implies that the posterior is not concentrated.

Extreme example: Model $y_i = f(\mathbf{x}_i) + \epsilon_i, \epsilon_i \stackrel{iid}{\sim} N(0, \frac{1}{4})$, Fourier basis, and a simple model misspecification: $y_i = 0 + \epsilon_i, \epsilon_i \stackrel{iid}{\sim} N(0, \frac{1}{4})$, $x_i \stackrel{iid}{\sim} U(-1, 1)$, **but then** set half of the data to $(0, 0)$.



Solution: Generalized (Safe-) Bayes with learning rate η [5]

$$\pi(\theta | x, \eta) := \frac{\ell_{\theta}(x)^{\eta} \pi(\theta)}{\int_{\Theta} \ell_{\theta}(x)^{\eta} \pi(\theta) d\theta}, \quad (1)$$

with $\ell_{\theta}(x)$ the likelihood and $\pi(\theta)$ the prior in case of parametric models and log loss.

- learn optimal η^* with the **Safe-Bayesian algorithm** [3]
- posterior concentrates on $P_{\hat{\theta}}$ with fast rates and mild condition ($\bar{\eta}$ -central condition) if η taken *small enough*, e.g. for GLMs [4]

Generalized Bayes II – Credal Sets

Credal Sets

In the IP literature, “generalized Bayes” typically refers to defining a set of priors

$$\Pi \subseteq \{\pi(\theta) \mid \pi(\cdot) \text{ a probability measure on } (\Theta, \sigma(\Theta))\} \quad (2)$$

with $\sigma(\cdot)$ an appropriate (σ -)algebra and Θ a (compact) parameter space. Inference then basically consists of updating Π to a set of posteriors.

Reformulation of Generalized Bayes I

Consider the numerator in the Bayes rule with learning rate η (equation 1). Note that $\ell_{\theta}(x)^{\eta} \pi(\theta) = \ell_{\theta}(x) [\pi(\theta) \ell_{\theta}(x)^{\eta-1}]$. This allows us to specify a set of priors given some base prior $\pi(\theta)$ as follows

$$\Pi_{\pi(\theta)} = \{\pi_{\nu}(\theta) \mid \tilde{\pi}(\theta) = \pi(\theta) \cdot \ell_{\theta}(x)^{\eta-1}, \eta \in (0, 1)\} \quad (3)$$

with normalization by $\pi_{\nu}(\theta) = \tilde{\pi}(\theta) / C_{\nu}, C_{\nu} = \int_{\Theta} \tilde{\pi}(\theta) d\theta$.

Bayes Theorem for Unbounded Priors [2]

Unnormalized versions of the prior functions in $\Pi_{\pi(\theta)}$ can be characterized by a point-wise upper and a (trivial) point-wise lower bound, namely $\pi(\theta) \cdot \ell_{\theta}(x)^{-1}$ and $\pi(\theta)$

- Sets of priors with these characteristics satisfy the requirements for *Bayes theorem for unbounded priors* [2, page 238]
- So a posterior credal set exists, see [1, chapter 2.3]

Thus, we can identify the posteriors with learning rates η (equation 1) with a posterior credal set.

Generalized Bayes I Through the Lens of Generalized Bayes II

Why does the learning rate allow concentration of the posterior under model misspecification?

- Assume the Safe-Bayesian algorithm finds the optimal η^*
- Consider our unnormalized prior for η^* , i.e. $\tilde{\pi}(\theta)^* = \pi(\theta) \cdot \ell_{\theta}(x)^{\eta^*-1}$
- $\tilde{\pi}(\theta)^*$ gives a **counterfactual Bayesian explanation** of Safe-Bayesian learning: *If we had specified the prior proportional to $\tilde{\pi}(\theta)$, we would have achieved concentration under model misspecification with regular Bayesian learning.*

- In this way, it conveys information on which parts of Θ are relevant to the non-concentration under misspecification.

Can this interpretation improve the Safe-Bayesian algorithm?

The Safe-Bayesian algorithm iterates both over a grid of η 's, and over each datapoint, i.e. the posterior has to be computed for each combination of η and each datapoint anew. Can we use our representation of the learning rates by a credal set somehow to speed up the search for the optimal η^* ?

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